2.5 FREQUENCY DISTRIBUTION

A ***frequency distribution*** is a convenient way of presenting a large mass of data in tabular form by grouping the data.

There are two types of frequency distributions, *viz. discrete* and *continuous.*

Discrete Frequency Distribution (or Ungrouped Data). A frequency distribution represented by a discrete variable is called a ***discrete frequency distribution.*** A discrete frequency distribution presents the data of a discrete variable in tabular form by listing all possible values that a discrete variable can take on, along with the corresponding frequencies.

If X is discrete variable that can take on values $x_1, x_2, \ldots, x_n$ with the corresponding frequencies $f_1, f_2, \ldots f_n$ then the frequency distribution of X is given by

X	:	x_1	x_2	.....	x_n
Frequency	:	f_1	f_2	.....	f_n

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For example, Table 2.5 shows a discrete frequency distribution.

TABLE 2.5. FREQUENCY DISTRIBUTION OF NUMBER OF CHILDREN

Number of Children	*Number of Families*
0	8
1	20
2	15
3	5
4	2
Total	50

Continuous Frequency Distribution. *A* frequency distribution represented by a continuous variable is called a *continuous frequency distribution. A continuous* frequency distribution presents the data of a continuous variable in tabular form by grouping the data into different classes and then recording the number of items that appear in each class.

For example, Table 2.6 shows the continuous frequency distribution of the heights of 50 students, recorded to the nearest centimeter :

TABLE 2.6. FREQUENCY DISTRIBUTION OF HEIGHTS OF 50 STUDENTS

Height (Centimeters)	*Number of Students (Frequency)*
155 – 156	2
157 – 158	4
159 – 160	9
161 – 162	14
163 – 164	10
165 – 166	7
167 – 168	3
169 – 170	1
	Total 50

Constructing a Discrete Frequency Distribution. *A* discrete frequency distribution can be constructed by preparing a table having three columns. In the first column, we place all possible values of the variable *(i.e.* the characteristic we are measuring) starting with the lowest and going up to the highest. In the second column labelled "Tally Marks", we indicate tally marks for each value, that is, we go through the list of original data and put a mark in the appropriate space for each value. To facilitate counting we, after drawing four marks, draw the fifth one across these to form what is often called a "gate". Finally, in the third column labelled "Frequency", we enter the total number of tally marks corresponding to each value of the variable. Each entry in the frequency column corresponding to a value represents the number of times that value is repeated in the data and is called its frequency.

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EXAMPLE 5. *The following data shows the marks obtained by 30 students of a class in Statistics :*

45	25	32	18	25	35	32	45	18	25
50	48	32	40	45	32	18	25	45	18
25	18	45	25	35	48	35	45	25	50

Construct a discrete frequency distribution for the above data.

SOLUTION. The discrete frequency distribution for the given data is shown in Table 2.7.

TABLE 2.7. DISCRETE FREQUENCY DISTRIBUTION FOR THE MARKS OF 30 STUDENTS

Marks	*Tally Marks*	*Frequency (No. of students)*
18	𝍸	5
25	𝍸 II	7
32	IIII	4
35	III	3
40	I	1
45	𝍸 I	6
48	II	2
50	II	2
		30

The above table shows that there are five students who have scored 18 marks, seven students who have scored 25 marks, and so on.

EXAMPLE 6. *The following data shows the number of children in 40 families.*

1	2	6	5	1	5	1	3	2	6
2	3	4	2	0	0	4	4	3	2
2	0	0	1	2	2	4	3	2	1
0	5	1	2	4	3	4	4	1	6

Construct a discrete frequency distribution for the above data.

SOLUTION. The discrete frequency distribution for the given data is shown in Table 2.8.

TABLE 2.8. DISCRETE FREQUENCY DISTRIBUTION

Number of Children	*Tally Marks*	*Number of Families (Frequency)*
0	𝍸	5
1	𝍸 II	7
2	𝍸 𝍸	10
3	𝍸	5
4	𝍸 II	7
5	III	3
6	III	3

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The Table 2.8 shows that there are five families having no children, seven families having only one child, ten families having exactly two children, and so on.

Some Useful Definitions. We shall now give some definitions which are relevant to a continuous frequency distribution, such as one shown in Table 2.9. Table 2.9 is a frequency distribution of the weights of a group of 60 students in a class, recorded to the nearest kilogram.

TABLE 2.9. FREQUENCY DISTRIBUTION FOR THE WEIGHTS OF 60 STUDENTS

Weight	*No. of Students*
30 – 34	3
35 – 39	5
40 – 44	12
45 – 49	18
50 – 54	14
55 – 59	6
60 – 64	2

Class Interval. An interval defining a class is called a ***class interval*** (or simply *class*). Each class interval is identified by the range of values assigned to the class. For example, in Table 2.9 the range of values of first class 30 – 34 is from 30 through 34 inclusive; the range of values of second class interval is from 35 through 39 inclusive; and so on.

Class Limits. The smallest and largest values that define a given class interval are referred to as its class limits. The smaller number is called the **lower class limit** and the larger number is called the **upper class limit.** For example, for the class interval 35 – 39, the class limits are 35 and 39. The smaller number, 35, is the lower class limit and the larger number, 39 is the upper class limit.

Types of Classes. There are two ways in which data can be grouped into different classes. These are (1) *Inclusive Method,* and (2) *Exclusive Method.*

1. ***Inclusive Method.*** Grouping of data into different classes such that both the lower and upper class limits of each class interval are included in that class is called grouping by "inclusive method". Here, the grouping of data into different classes is done in such a way that the upper class limit of one class does not coincide with the lower class limit of the next class. For example, the marks obtained by students in a class may be grouped as 0 – 9, 10 – 19, 20 – 29, ... , where the class interval 0 – 9 includes all values from 0 to 9 (both inclusive), the class interval 10 – 19 includes all values from 10 to 19, and so on. Thus, for instance, if a student has secured 10 marks, he is included in the class 10 – 19.
2. ***Exclusive Method.*** Grouping of data into different classes such that the upper class limit of one class is same as the lower class limit of the next class is called grouping by "exclusive method". Here, it should be understood that the upper class limit of any class is excluded from that class. For example, if the marks obtained by students in a class are grouped as 0 – 10, 10 – 20, 20 – 30, ... , then the class interval 0 – 10 includes all values which are greater than or equal to 0 but less than 10, the class

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interval 10 – 20 includes all values which are greater than or equal to 10 but less than 20, and so on. Thus if a student has secured 10 marks, he will be included in the class 10 – 20.

Class Boundaries. The **class boundaries** are the values halfway between the upper class limit of one class and the lower class limit of the next class. Each class has an upper and lower class boundaries. The class boundaries are obtained so as to fill the gap that might exist between classes.

If d is the gap between the upper class limit of one class and the lower class limit of next class, the class boundaries for any class are given by

$$\text{Lower class boundary} = \text{lower class limit} - \frac{d}{2}$$

$$\text{Upper class boundary} = \text{upper class limit} + \frac{d}{2}$$

For the frequency distribution given in Table 2.9 the gap between the upper class limit of one class and the lower class limit of next class is $d = 1$. Dividing it by 2 we get 1/2 or 0.5. Subtracting this value from all lower class limits and adding the same to all upper class limits, we obtain the class boundaries for each class as shown in Table 2.10.

TABLE 2.10. FREQUENCY DISTRIBUTION FOR THE WEIGHTS OF 60 STUDENTS

Class Interval	*Class Boundaries*	*No. of Students*
30 – 34	29.5 – 34.5	3
35 – 39	34.5 – 39.5	5
40 – 44	39.5 – 44.5	12
45 – 49	44.5 – 49.5	18
50 – 54	49.5 – 54.5	14
55 – 59	54.5 – 59.5	6
60 – 64	59.5 – 64.5	2

Notice that class boundaries are always carried out to one more decimal place than the recorded observation. The class limits are expressed in the units used for measuring. The class boundaries, however, are expressed in units that are smaller than the units used in the measurements. For example, if the class limits are expressed in whole number units, then the class boundaries are expressed in tenths of a unit. Or, if the limits are expressed in tenths of a unit, then the boundaries are expressed in hundredths of a unit, and so on.

Class Width. The difference between the upper and lower class boundaries of a class interval is called the *class width* (or *class size).*

In Table 2.10 all class intervals are of equal width. In fact, the width of each class is 5.

Class Frequency. The number of observations that fall in a particular class is called the ***class frequency*** and is denoted by the letter 'f'. The class frequency can also be expressed as a fraction or a percentage of the total frequency. We refer to this value as the ***relative frequency*** of a class. A relative frequency distribution presents frequencies in terms of fractions or percentages. The relative frequency distribution for the data in Table 2.9 is given in Table 2.11.

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TABLE 2.11. RELATIVE FREQUENCY DISTRIBUTION

Class Interval	*Class Boundaries*	*Class Mark*	*Frequency*	*Relative Frequency*
30 – 34	29.5 – 34.5	32	3	3/60
35 – 39	34.5 – 39.5	37	5	5/60
40 – 44	39.5 – 44.5	42	12	12/60
45 – 49	44.5 – 49.5	47	18	18/60
50 – 54	49.5 – 54.5	52	14	14/60
55 – 59	54.5 – 59.5	57	6	6/60
60 – 64	59.5 – 64.5	62	2	2/60

Class Marks. The value of the midpoint of the class boundaries or class limits of a class interval is called the **class mark** or **class midpoint.** Thus

$$\text{Class mark} = \frac{\text{lower class limit} + \text{upper class limit}}{2}$$

$$= \frac{\text{lower class boundary} + \text{upper class boundary}}{2}$$

For data in Table 2.11 the class marks are 32, 37, 42, 47, 52, 57 and 62.

NOTE. It may be noted that the class mark is used as the representative value of a given class. For example, for the first class interval 30 – 34, the class mark is 37 which is used as a representative value for all data included in that class.

Open-ended Class Interval. A class interval with an unspecified upper or lower class limit is called an **open-ended class interval.**

For example, classes such as salary less than Rs. 10,000; age above 65 years; marks less than 10, etc. are "open-ended" classes since one of the class limits is not specified in each of them.

Cumulative Frequency Distribution. The cumulative frequency can be classified into two types: *(i) "less than" cumulative frequency, (ii) "more than" cumulative frequency.*

(i) "Less than" Cumulative Frequency. The number of observations which are less than the upper class boundary of a given class interval is called the ***"less than" cumulative frequency*** of that class. Thus the "less than" cumulative frequency of a given class interval is the sum of the frequency of the given class and the frequencies of all classes having a class mark less than that of the given class.

(ii) "More than" Cumulative Frequency. The number of observations which are greater than the lower class boundary of a given class interval is called the ***"more than" cumulative frequency*** of that class. Thus the "more than" cumulative frequency of a given class interval is the sum of the frequency of the gives class and the frequencies of all classes having a class mark greater than that of the given class.

A table listing cumulative frequencies is called a **cumulative frequency distribution.** The cumulative frequency distribution for the data given in Table 2.9 is shown in Table 2.12.

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TABLE 2.12. CUMULATIVE FREQUENCY DISTRIBUTION

Class Interval	*Class Boundaries*	*Class Mark*	*Frequency*	*Cum.Freq. (Less than)*	*Cum.Freq. (More than)*
30 – 34	29.5 – 34.5	32	3	3	57 + 3 = 60
35 – 39	34.5 – 39.5	37	5	3 + 5 = 8	52 + 5 = 57
40 – 44	39.5 – 44.5	42	12	8 + 12 = 20	40 + 12 = 52
45 – 49	44.5 – 49.5	47	18	20 + 18 = 38	22 + 18 = 40
50 – 54	49.5 – 54.5	52	14	38 + 14 = 52	8 + 14 = 22
55 – 59	54.5 – 59.5	57	6	52 + 6 = 58	2 + 6 = 8
60 – 64	59.5 – 64.5	62	2	58 + 2 = 60	2

EXAMPLE 7. *Convert the following into an ordinary frequency distribution :*

5 students get less than 3 marks ;
12 students get less than 6 marks ;
25 students get less than 9 marks ;
33 students get less than 12 marks. ***[Delhi Univ. B.Com. 2001]***

SOLUTION. The required ordinary frquency distribution is shown in Table 2.13.

TABLE 2.13. FREQUENCY DISTRIBUTION

Marks	*No. of Students (f)*
0 – 3	5
3 – 6	12 – 5 = 7
6 – 9	25 – 12 = 13
9 – 12	33 – 25 = 8

EXAMPLE 8. *Convert the following distribution into "more than" frequency distribution :*

Weekly wages "less than" (Rs.)	:	*20*	*40*	*60*	*80*	*100*
No. of workers	:	*41*	*92*	*156*	*194*	*201*

[Delhi Univ. B.Com.1979]

SOLUTION. To obtain the 'more than' frequency distribution, we shall first convert it into an ordinary frequency distribution as shown in Table 2.14 (*a*).

The required 'more than' frequency distribution can now easily be obtained from Table 2.14 (*a*) as shown in Table 2.14 (*b*).

TABLE 2.14 (*a*)
ORDINARY FREQUENCY DISTRIBUTION

Weekly Wages (Rs.)	*No. of workers*
0 – 20	41
20 – 40	92 – 41 = 51
40 – 60	156 – 92 = 64
60 – 80	194 – 156 = 38
80 – 100	201 – 194 = 7

TABLE 2.14 (*b*)
'MORE THAN' FREQUENCY DISTRIBUTION

Weekly wages more than (Rs.)	*No. of workers*
0	160 + 41 = 201
20	109 + 51 = 160
40	45 + 64 = 109
60	7 + 38 = 45
80	7

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Constructing a Continuous Frequency Distribution

We now outline the procedure for constructing a continuous frequency distribution.

Step 1. ***To determine the number of classes.*** *The first step in the construction of a continuous frequency distribution is to determine the number of classes that will be used to group the data. There are no strict rules governing how many classes to use for a given data, but, in general, the number of classes varies from a minimum of 5 to a maximum of 18. In fact the number of classes depends on the number of data points and the range of the data collected. Some statisticians use a formula known as* ***'Sturges' Rule*** *to determine the necessary number of classes :*

$$k = 1 + 3.3 \log_{10} n$$

where n is the number of items in the given data and k is the number of required classes.

It is worth mentioning here that the classes must be selected in such a way so as to conform to following rules :

(*i*) All data items from the smallest to the largest must be included, and

(*ii*) Each item must be assigned to one and only one class.

Step 2. ***To determine the width of each class.*** *Once we determine the number of classes to use for a given distribution, the next step is to determine the width for each class. In practice, it is desirable to have equal widths for all class intervals. If the classes were unequal and the width of the intervals differed among classes, then we would have a distribution that is much more difficult to interpret than the one with equal intervals. Generally speaking, if we subtract the smallest value in data from the largest value and divide the result by total number of classes, we will get a number, rounded off if necessary, which can be used as the width for each class. That is,*

$$\text{Class width} = \frac{\text{largest value} - \text{smallest value}}{\text{total number of classes}} \quad \text{(and then round up)}$$

$$= \frac{\text{range}}{\text{total number of classes}}$$

Step3. ***To determine the frequency of each class.*** *Finally, we count the number of observations falling in each class and record the appropriate number in the frequency column. It is advisable to use the method of "Tally Marks" to find the frequency of each class.*

These three steps enable us to arrange the data in both tabular and graphic form.

EXAMPLE 9. *The following data represent the marks obtained by 50 students in an examination :*

29	28	16	23	22	10	8	15	16	23
29	29	22	15	24	9	5	4	15	24
21	24	16	14	21	13	6	2	23	22
24	21	27	13	20	8	4	3	24	20
22	18	23	23	17	7	6	1	16	17

Construct a frequency distribution for the above data.

SOLUTION. We first decide on the number of classes into which data are to be grouped. Let us arbitrarily choose 6 class intervals. To determine the width of each class, we divide the

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range by the number of classes. The largest value is 29 and the smallest value is 1.

$$\therefore \quad \text{Range} = \text{largest value} - \text{smallest value} = 29 - 1 = 28$$

$$\Rightarrow \quad \text{Class width} = \frac{\text{range}}{\text{no. of classes}} = \frac{28}{6} = 4.67 \quad \text{or} \quad 5 \text{ when rounded}$$

TABLE 2.15. FREQUENCY DISTRIBUTION OF MARKS

Class Interval	*Tally Marks*	*Frequency*
1 – 5	𝍸 I	6
6 – 10	𝍸 II	7
11 – 15	𝍸 II	7
16 – 20	𝍸 III	8
21 – 25	𝍸 𝍸 𝍸 II	17
26 – 30	𝍸	5

Thus the class width for each class interval is 5. Taking the first class interval as 1 – 5 and using tally marks, we construct the required frequency distribution as shown in Table 2.15.

EXAMPLE 10. *The heights in cms. of 30 persons are given below :*

133	*125*	*137*	*129*	*130*	*130*	*131*	*125*	*137*	*147*
128	*127*	*147*	*141*	*148*	*149*	*145*	*148*	*139*	*125*
145	*134*	*129*	*145*	*127*	*147*	*132*	*128*	*130*	*131*

Prepare a frequency distribution for the above data.

SOLUTION. We first decide on the number of classes into which data are to be grouped. Let us arbitrarily choose 9 classes. To determine the width of each class, we divide the range by the number of classes. The largest value is 149 and the smallest value is 125.

$$\therefore \quad \text{Range} = \text{largest value} - \text{smallest value} = 149 - 125 = 24$$

$$\Rightarrow \quad \text{Class width} = \frac{\text{range}}{\text{no. of classes}} = \frac{24}{9} = 2.67 \quad \text{or} \quad 3 \text{ when rounded}$$

Thus the class width for each class is 3. Taking the first class interval as 125 – 127 and using tally marks, we get the required frequency distribution as shown in Table 2.16.

TABLE 2.16. FREQUENCY DISTRIBUTION OF HEIGHTS

Class Interval	*Tally Marks*	*Frequency*
125 – 127	𝍸	5
128 – 130	𝍸 II	7
131 – 133	IIII	4
134 – 136	I	1
137 – 139	III	3
140 – 142	I	1
143 – 145	III	3
146 – 148	IIII	4
149 – 151	II	2
		30

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AMPLE 11. *Construct a frequency distribution of the marks obtained by 50 students in Economics as given below :*

42	53	65	63	61	47	58	60	64	45
55	57	82	42	39	51	65	55	33	70
50	52	53	45	45	25	36	59	63	39
65	30	45	35	49	15	54	48	64	26
75	20	42	40	41	55	52	46	35	18

Take the first class interval as 10 – 20) **[*Delhi Univ. B.Com.1983*]**

LUTION. The required frequency distribution is shown in Table 2.17.

TABLE 2.17. FREQUENCY DISTRIBUTION OF MARKS

Class Interval	*Tally Marks*	*Frequency*
10 – 20	II	2
20 – 30	III	3
30 – 40	𝍸 II	7
40 – 50	𝍸 𝍸 III	13
50 – 60	𝍸 𝍸 III	13
60 – 70	𝍸 IIII	9
70 – 80	II	2
80 – 90	I	1
		50

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